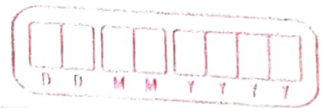


Assignment



Q1) a) Decimal - 143.75

$$143.75 \Rightarrow 2 \overline{) 143}$$

$$2 \overline{) 71} - 1$$

$$2 \overline{) 35} - 1$$

$$2 \overline{) 17} - 1$$

$$2 \overline{) 8} - 1$$

$$2 \overline{) 4} - 0$$

$$2 \overline{) 2} - 0$$

$$\underline{1} - 0$$

$$0.75 \times 2 = 1.50 - 1$$

$$0.50 \times 2 = 1.00 - 1$$

$$= \underline{\underline{0.11}}$$

$$= \underline{\underline{10001111}}$$

$$\text{Binary} \Rightarrow \underline{\underline{10001111.11}}_{(2)}$$

Octal - 143.75
(8)

$$8 \overline{) 143}$$

$$8 \overline{) 17} - 7 = \underline{\underline{217}}$$

$$8 \overline{) 2} - 1$$

$$0 - 2$$

$$0.75 \times 8 = \underline{\underline{6}}$$

$$\text{Octal} \Rightarrow \underline{\underline{217.6}}_{(8)}$$

Hexadecimal
(16)

$$16 \overline{) 143}$$

$$16 \overline{) 8} - 15 = \underline{\underline{815}}$$

$$0 - 8$$

$$0.75 \times 16 = \underline{\underline{12}}$$

$$\therefore \underline{\underline{815.12}}$$

$$\text{Hexadecimal} \Rightarrow \underline{\underline{8F.C}}_{(16)}$$

b) Binary - $11011.101_{(2)}$

$$\begin{aligned}\text{Decimal} &\Rightarrow 1 \times 2^4 + 1 \times 2^3 + 0 \times 2^2 + 1 \times 2^1 + 1 \times 2^0 \\ &= 1 + 2 + 0 + 8 + 16 = \underline{27}\end{aligned}$$

$$\begin{aligned}101 &\Rightarrow 1 \times 2^{-1} + 0 \times 2^{-2} + 1 \times 2^{-3} = 0.5 + 0 + 0.125 = 0.625 \\ &= \underline{27.625_{(10)}}$$

$$\begin{aligned}\text{Octal} &- \underline{011011.101} \\ &\quad \downarrow \quad \downarrow \quad \downarrow \\ &\Rightarrow 35.5\end{aligned}$$

$$\therefore \text{Octal} = \underline{35.5_{(8)}}$$

Hexadecimal - 11011.101

$$\begin{array}{ccc}\underline{0001} & \underline{1011} & \underline{1010} \\ 1 & 11 & 10 \\ \downarrow & \downarrow & \downarrow \\ 1 & B & A\end{array}$$

$$\therefore \underline{1B.A_{(16)}}$$

c) Octal - 453₍₈₎

$$\begin{aligned}\text{Decimal} &= 4 \times 8^2 + 5 \times 8^1 + 3 \times 8^0 \\ &= 256 + 40 + 3 \\ &= \underline{\underline{299}}\end{aligned}$$

$$\begin{aligned}\text{Binary} - 453 &= \overset{4}{100}, \overset{5}{101}, \overset{3}{011} \\ &= \underline{\underline{100101011}}_{(2)}\end{aligned}$$

$$\begin{aligned}\text{Hexadecimal} - 453 &= \underbrace{1001}_{12} \underbrace{0101}_{15} \underbrace{11}_{3} \\ &= \underline{\underline{12B}}_{(16)}\end{aligned}$$

d) Hexadecimal - 1F.C₍₁₆₎

$$\begin{aligned}\text{Decimal} &= 1 \times 16^1 + F \times 16^0 + C \times 16^{-1} \\ &= 16 + 15 + 12 \times 0.0625 \Rightarrow (0.75) \\ &= \underline{\underline{31.75}}\end{aligned}$$

$$\begin{aligned}\text{Binary} = 1F.C &\rightarrow \underbrace{0001}_1 \underbrace{1111}_{15(F)} \underbrace{1100}_{12(C)} \Rightarrow \underline{\underline{1111.11}}_{(2)}\end{aligned}$$

$$\begin{aligned}\text{Octal} = 1F.C &\rightarrow \underbrace{01}_3 \underbrace{1111}_7 \underbrace{110}_6 \Rightarrow \underline{\underline{37.6}}_{(8)}\end{aligned}$$

Q2) Find Equivalents

$$a) 324_{(5)} = ?_{(10)} = 3 \times 5^2 + 2 \times 5^1 + 4 \times 5^0 \\ = 75 + 10 + 4 = \underline{89}_{(10)}$$

$$b) 32.23_{(4)} = ?_{(10)} = 3 \times 4^1 + 2 \times 4^0 + 2 \times 4^{-1} + 3 \times 4^{-2} \\ = 12 + 2 + 0.5 + 0.1875 \\ = \underline{14.6875}_{(10)}$$

$$c) 231_{(7)} \Rightarrow ?_{(17)} = \begin{array}{r} 7 \overline{) 231} \\ 7 \overline{) 33} - 0 \\ 7 \overline{) 4} - 5 \\ 0 - 4 \end{array} = \underline{450}_{(17)}$$

$$d) 189_{(10)} = ?_{(6)} = \begin{array}{r} 6 \overline{) 189} \\ 6 \overline{) 31} - 3 \\ 6 \overline{) 5} - 1 \\ 0 - 5 \end{array} = \underline{513}_{(6)}$$

Q3) Find the Radix x :

$$a) (312/20)_x = (13.1)_x \Rightarrow \frac{3x^2 + x + 2}{2x} = x + 3 + \frac{1}{x}$$

$$312_x = 3x^2 + 1x^1 + 2x^0 = 3x^2 + x + 2$$

$$20_x = 2x^1 + 0x^0 = 2x$$

$$13.1_x = 1x^1 + 3x^0 + 1x^{-1} \Rightarrow x + 3 + \frac{1}{x}$$

$$3x^2 + x + 2 = 2x \left(x + 3 + \frac{1}{x} \right)$$

$$3x^2 + x + 2 = 2x^2 + 6x + 2$$

$$3x^2 - 2x^2 - 6x + x + 2 - 2 = 0$$

$$x^2 - 5x = 0 \Rightarrow \underline{x = 5}$$

$$x(x - 5) = 0$$

$$b) 264r + 136r = 433r$$

$$264r = 2r^2 + 6r + 4 \quad (2r^2 + 6r + 4) + (1r^2 + 3r + 6) = 4r^2 + 3r + 3$$

$$136r = r^2 + 3r + 6 \quad 3r^2 + 9r + 10 = 4r^2 + 3r + 3$$

$$433r = 4r^2 + 3r + 3 \quad r^2 - 6r - 7 = 0$$

$$(r-7)(r+1) = 0$$

$$r = 7$$

$$c) 221505_{(8)} = 12345_r$$

$$\begin{aligned} 221505_{(8)} &= 2(8)^5 + 2(8)^4 + 1(8)^3 + 5(8)^2 + 0(8)^1 + 5(8)^0 \\ &= 65536 + 8192 + 512 + 320 + 5 \\ &= \underline{74565} \end{aligned}$$

$$\begin{aligned} 12345_r &= 1r^4 + 2r^3 + 3r^2 + 4r^1 + 5r^0 \\ &= 16^4 + 2(16)^3 + 3(16)^2 + 4(16)^1 + 5(16)^0 \\ &= 65536 + 8192 + 768 + 64 + 5 \\ &= \underline{74565} \end{aligned}$$

$$r = 16$$

Q4) Addition

$$a) 1101_2 + 1101111_2$$

$$\begin{array}{r} 1101111 \\ 1101 \\ \hline 11101100 \end{array}$$

$$b) 77652_{(8)} + 76543_{(8)}$$

$$\begin{array}{r} 77652 \\ 76543 \\ \hline 176395_{(8)} \end{array}$$

$$578A_{16} + ABDE_{16}$$

$$\begin{array}{r} \overset{1}{5} \quad \overset{1}{7} \quad \overset{1}{8} \quad A \\ 1 \quad A \quad B \quad D \quad E \\ \hline 1 \quad 16 \quad 19 \quad 22 \quad 24 \end{array}$$

$$24 - 16 = 8$$

$$22 - 16 = 6$$

$$19 - 16 = 3$$

$$16 - 16 = 0$$

1

$$\Rightarrow \underline{\underline{10368}}_{(16)}$$

$$234_{(6)} + 315_{(6)}$$

$$\begin{array}{r} \overset{1}{2} \quad \overset{1}{3} \quad \overset{1}{4} \Rightarrow 2(6)^2 + 3(6) + 4 = \underline{\underline{94}} \end{array}$$

$$\begin{array}{r} \overset{1}{3} \quad \overset{1}{1} \quad \overset{1}{5} \Rightarrow 3(6)^2 + 1(6) + 5 = \underline{\underline{119}} \end{array}$$

$$\begin{array}{r} 8 \quad 5 \quad 9 \end{array}$$

$$\begin{array}{r} \overset{1}{2} \quad \overset{1}{3} \quad \overset{1}{4} = 9 = 1 * 6 * 3 = \end{array}$$

$$\begin{array}{r} 3 \quad 1 \quad 5 \\ \hline \end{array}$$

$$\underline{\underline{553}}$$

$$\therefore 553_{(6)}$$

Binary $\rightarrow$ Gray Code

$$10101_{(2)} \rightarrow 1, 1+0=1, 0+1=1, 1+0=1, 0+1=1$$

$$\therefore \underline{\underline{11111}}$$

b) $01110_{(2)} \Rightarrow 0, 0+1=1, 1+1=0, 1+1=0, 1+0=1.$

$\therefore \underline{\underline{01001}}$

Q 6) ~~Gray~~ Code $\rightarrow$ Binary

a) $01101 = 0, 0+1=1, 1+1=0, 0+0=0, 0+1=1$

$\therefore \underline{\underline{01001_2}}$

b) $10101 = 1, 1+0=1, 1+1=0, 0+0=0, 0+1=1$

$\therefore \underline{\underline{11001_2}}$